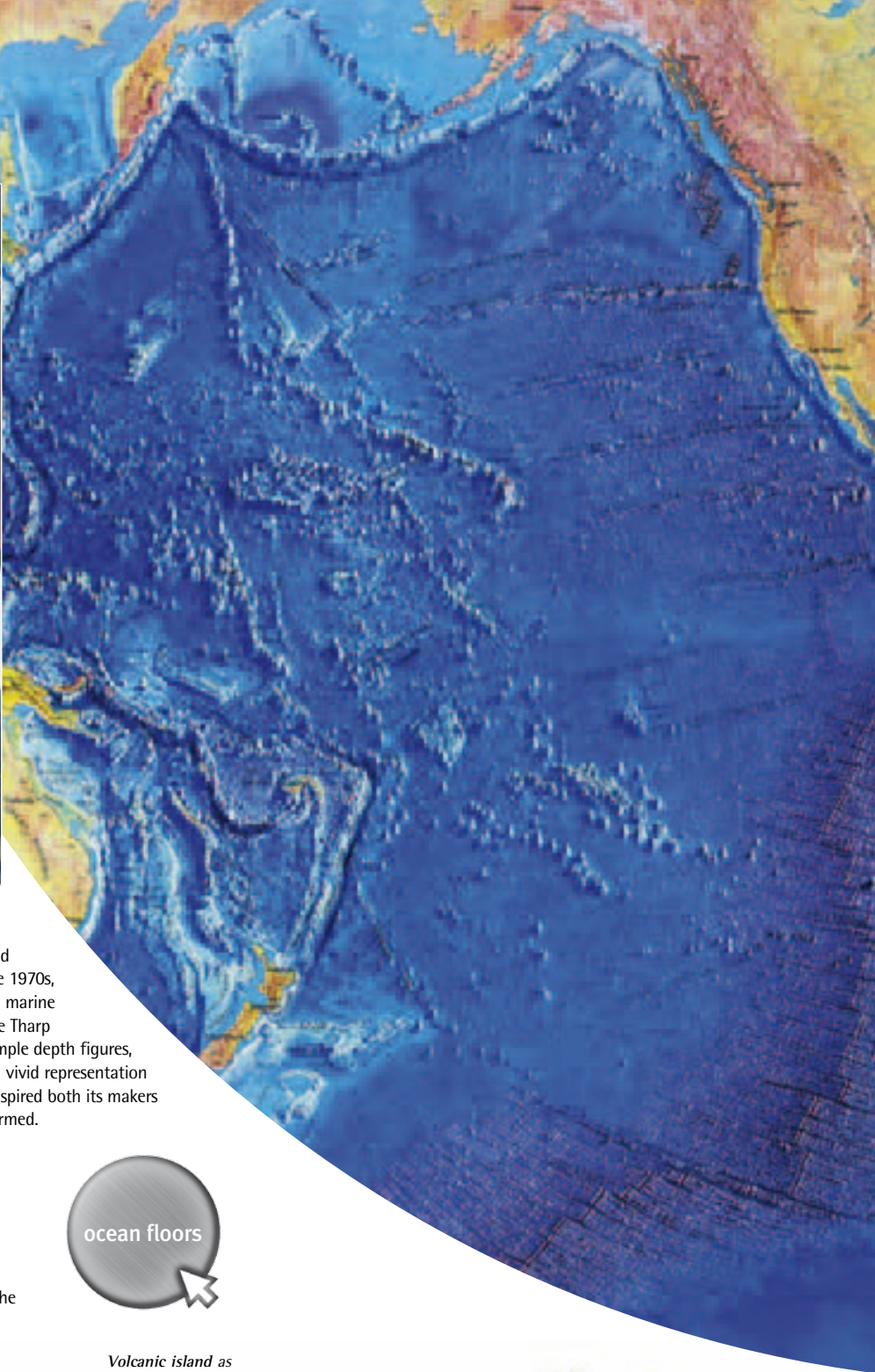




AN OCEAN FLOOR MAP ▲

The development of echo-sounding in the 1920s enabled ships to record ocean depths easily and quickly. As the data built up, it started to create a picture of the ocean floors. This was made a reality in the 1970s, with the publication of the first world ocean floor map by American marine geologist Bruce Heezen (1924–1977) and his research assistant Marie Tharp (1920–2006), seen above. The map (right) was put together using simple depth figures, so a lot of the detail was based on educated guesswork. But it was a vivid representation of the ocean floors, and the patterns and features that it revealed inspired both its makers and other scientists to discover more about how the oceans were formed.



▼ MORE REALISTIC HEIGHTS AND SLOPES

In reality, oceanic volcanoes are broad domes, and the gradients at the edges of shallow coastal seas are so shallow that, if these gradients were on land, you could ride a bicycle up them. Even deep ocean trenches are broad troughs rather than steep-sided chasms. This illustration shows a more realistic view of the scene depicted opposite.

Volcanic island as it is in reality

